

PROJECT 1ST

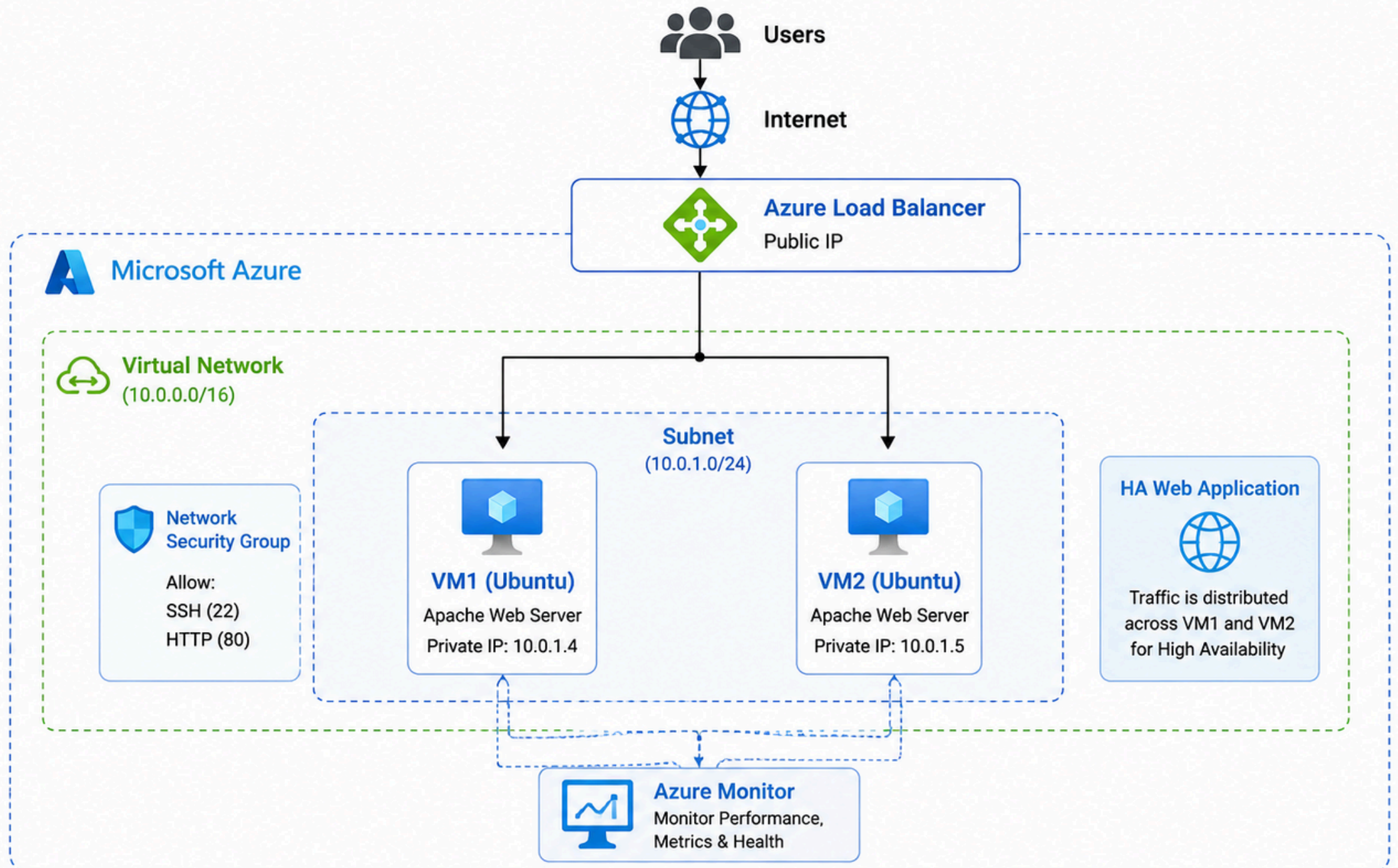
Cloud Infrastructure

Deployment On

Microsoft Azure

CLOUD INFRASTRUCTURE ARCHITECTURE

High Availability Web Application on Microsoft Azure



PROJECT 1ST

Cloud Infrastructure Deployment On Microsoft Azure

KEY AZURE SERVICES USED



Virtual Machines

Deploy and manage scalable virtual machines in Azure.



Virtual Network

Create isolated networks and secure communication.



Network Security Group

Control inbound and outbound traffic to secure resources.



Load Balancer

Distribute incoming traffic across multiple VMs.



Azure Monitor

Monitor performance and health of resources.

Project Snapshots

Virtual machine
Linux Operating
system

using Load Balancer
to Distributed the
Traffic

Home > Compute infrastructure | Virtual machines

cloudproject2
Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search

Connect Start Restart Stop Hibernate Capture

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
- Networking
- Settings

Essentials

Resource group (move) : [Project-1](#)

Status : Running

Location : Korea Central (Zone 1)

Subscription (move) : [Azure for Students](#)

Subscription ID : 10b13e03-e217-496d-975c-d1efc433d9a1

Availability zone : 1

Tags (edit) : [Add tags](#)

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Load balancing and content delivery | Load balancers

project-lb
Load balancer

Run diagnostic for my Load Balancer How to avoid downtime due to load balancer outages Show me alerts for this load balancer

Search

Move Delete Refresh Give feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Essentials

Resource group (move) :	Project-1	Backend pool :	project1 (2 virtual machines)
Location (move) :	Korea Central	Load balancing rule :	project1 (Tcp/80)
Subscription (move) :	Azure for Students	Health probe :	project1 (Tcp:80)
Subscription ID :	10b13e03-e217-496d-975c-d1efc433d9a1	Inbound NAT rules :	-
SKU :	Standard	Frontend IP address :	10.1.0.6
Tier :	Regional		
Tags (edit) :	Add tags		

Configure high availability and scalability for your applications

Create highly-available and scalable applications in minutes by using built-in load balancing for cloud services and virtual machines. Az Load Balancer supports TCP/UDP-based protocols and protocols used for real-time voice and video messaging applications. [Learn more](#)

Project Snapshots

The screenshot shows the Microsoft Azure portal interface. At the top, the 'Microsoft Azure' header is visible with a search bar. The breadcrumb trail indicates the location: 'Home > Network foundation | Virtual networks'. The main heading is 'cloudproject-vnet' with the subtitle 'Virtual network'. Below this, there are buttons for 'Analyze traffic within this network' and 'Diagnose issues with this virtual network'. A search bar is present on the left. The left sidebar contains a list of navigation options: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, and Monitoring. The main content area shows the 'Essentials' section with the following details: Resource group (move) : [Project-1](#), Location (move) : Korea Central, Subscription (move) : [Azure for Students](#), and Subscription ID : 10b13e03-e217-496d-975c-d1efc433d9a1. At the bottom, there is a 'Tags (edit)' section with a link to 'Add tags'.

Cloud Virtual network

Using monitoring to Analyze the logs and matrices

The screenshot shows the Microsoft Azure Monitor | Overview page. The header includes the 'Monitor | Overview' title and a search bar. Below the header, there are buttons for 'Refresh', 'Help us improve', and 'Switch to legacy overview'. The left sidebar contains a list of navigation options: Overview (selected), Activity log, Alerts, Metrics, Observability agents (preview), Issues, Logs, Change Analysis, Service health, Workbooks, Dashboards with Grafana, Insights, Managed Services, Settings, and Support + Troubleshooting. The main content area shows the 'Top actions' section with a 'Service group' card that includes a description and a 'Create service group' button. Below this is the 'Monitoring status' section, which includes 'Azure service incidents' (showing 'No service incidents for the selected subscriptions' and '0 resolved incidents in the past 24 hours') and 'Issues in the last 24 hours (trend)' (showing a table with columns for Total issues, Critical, Error, Warning, Informational, and Verbose, all with values of 0).

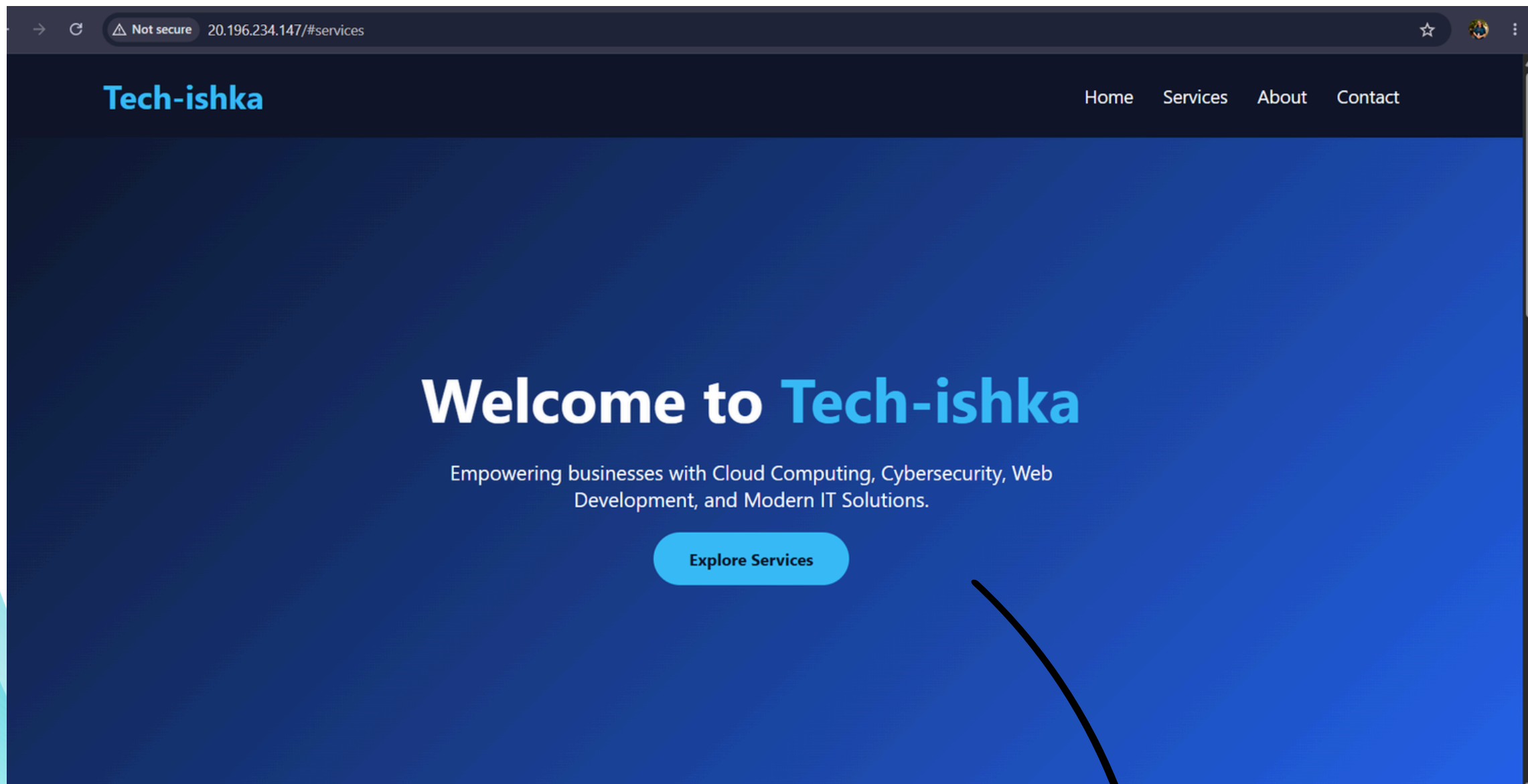
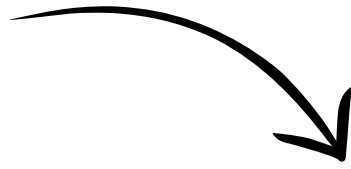
Project Snapshots

```
azure-tanishka@cloudproject  ×  +  v
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-08-06 04:28:59 UTC; 1min 24s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 17152 (apache2)
    Tasks: 55 (limit: 9439)
  Memory: 5.9M (peak: 6.1M)
    CPU: 45ms
   CGroup: /system.slice/apache2.service
           └─17152 /usr/sbin/apache2 -k start
             └─17155 /usr/sbin/apache2 -k start
               └─17156 /usr/sbin/apache2 -k start

Aug 06 04:28:59 cloudproject2 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 06 04:28:59 cloudproject2 systemd[1]: Started apache2.service - The Apache HTTP Server.
~
~
~
~
~
~
~
```

**Command line interface to upload Website
designing code and for install Apache on our
Linux OS**

Project Snapshots



Here's the final output